

ESAT CAMP MOCK TEST C

Physics

EC-PHY-C

PHYSICS

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Physics

- 1 A magnet rests on a top-pan balance. A straight horizontal wire of length 4.0 cm is held between the magnet's poles and carries current I . The balance reading increases linearly from 500.0 g at $I = 0$ to 504.0 g at $I = 2.0\text{ A}$. What is the magnitude of the magnetic field strength between the poles? Take $g = 10\text{ N kg}^{-1}$.
- A 0.0050 T
 - B 0.050 T
 - C 0.50 T
 - D 2.0 T
 - E 5.0 T
 - F 50 T

- 2 A block of mass 3.0 kg rests on a smooth, horizontal table. It is connected by two light, inextensible strings passing over smooth, frictionless pulleys at opposite ends of the table to two hanging masses of 5.0 kg and 2.0 kg, as shown in the diagram.

The system is released from rest.

What is the tension T_1 in the string attached to the 5.0 kg mass and the tension T_2 in the string attached to the 2.0 kg mass while the blocks are moving?

[Take the acceleration due to gravity to be $g = 10 \text{ m s}^{-2}$.]

	T_1 / N	T_2 / N
A	15	6
B	15	20
C	35	20
D	35	26
E	50	20
F	50	26
G	65	14
H	65	20

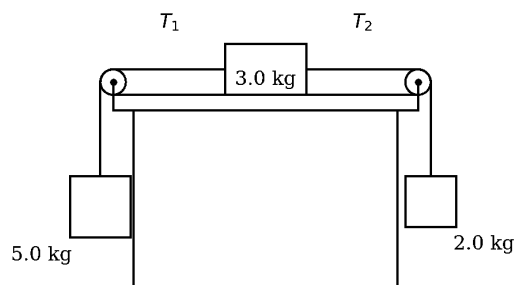


Diagram not to scale

- 3 A laser beam reflects off a plane mirror M_1 and then off a second plane mirror M_2 . The direction of the beam incident on M_1 is fixed. M_1 is rotated by an angle α in a clockwise direction.

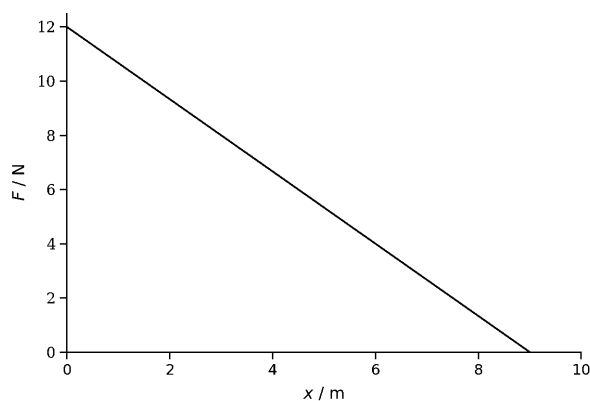
To ensure the final beam reflected from M_2 maintains a constant direction, M_2 must be rotated by:

- A α in the clockwise direction
- B α in the anticlockwise direction
- C 2α in the clockwise direction
- D 2α in the anticlockwise direction
- E $\frac{\alpha}{2}$ in the clockwise direction

- 4 A block of mass 2.0 kg is initially moving at a speed of 4.0 m s^{-1} along a smooth horizontal surface in the positive x -direction.

A single horizontal force F acts on the block in the direction of its motion. The graph shows how F varies with the displacement x of the block.

What is the speed of the block when it reaches $x = 6.0 \text{ m}$?



- A 6.0 m s^{-1}
- B 6.9 m s^{-1}
- C 7.2 m s^{-1}
- D 8.0 m s^{-1}
- E 8.4 m s^{-1}
- F 9.4 m s^{-1}

- 5 A uniform rectangular block of width 1.0 m and height 2.0 m rests on a rough horizontal surface. The coefficient of static friction between the block and the surface is μ . A horizontal force is applied to the top edge of the block. As the force is gradually increased, the block begins to tip over without sliding. Which inequality defines the possible values of μ ?

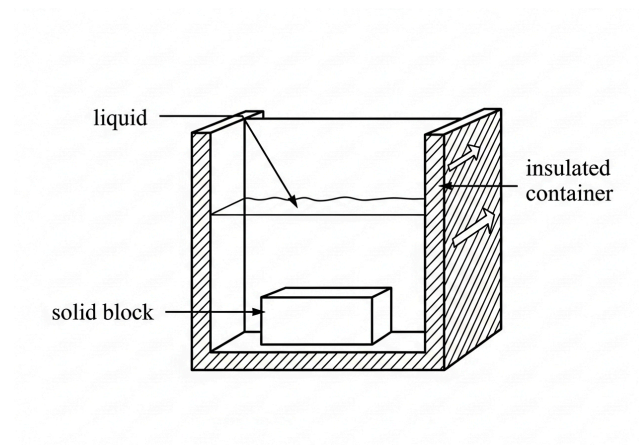
- A $\mu < 0.25$
- B $\mu > 0.25$
- C $\mu > 0.50$
- D $\mu > 1.00$
- E $\mu > 2.00$

- 6 A hot solid block is placed into a cold liquid in an insulated container.

The mass of the liquid is twice the mass of the block. The specific heat capacity of the liquid is three times the specific heat capacity of the block.

The temperature of the block decreases by ΔT .

Assuming no thermal energy is transferred to the surroundings, what is the temperature increase of the liquid?



- A $\frac{\Delta T}{6}$
- B $\frac{\Delta T}{5}$
- C $\frac{\Delta T}{3}$
- D $\frac{\Delta T}{2}$
- E $5\Delta T$
- F $6\Delta T$

- 7 The activity of a radioactive source is measured. Its activity after 1 hour is 8 times its activity after 4 hours.
Its activity after 2 hours is 120 Bq.
What was the initial activity of the source in Bq?
- A** 30
B 120
C 240
D 480
E 960
F 1920
- 8 A sealed syringe with a freely moving, frictionless piston contains a fixed mass of an ideal gas.
At a depth of 30 m below the surface of a lake, the volume of the gas in the syringe is 80 cm^3 .
When the syringe is moved to a depth of 10 m below the surface, the volume of the gas increases to 120 cm^3 .
The temperature of the water is uniform throughout the lake, and the gas remains in thermal equilibrium with the water.
What is the volume of the gas in the syringe at the surface of the lake?
- A** 135 cm^3
B 140 cm^3
C 150 cm^3
D 160 cm^3
E 180 cm^3
F 200 cm^3
G 240 cm^3
H 360 cm^3
- 9 Four identical sealed copper containers, P, Q, R, and S, are filled with equal masses of water at 80°C . The containers differ in their outer surface finish and the environment in which they are placed. The temperature of the surroundings in all cases is 20°C . | Container | Surface finish | Environment | | :--- | :--- | :--- | | P | Dull black | Air | | Q | Shiny silver | Air | | R | Dull black | Vacuum | | S | Shiny silver | Vacuum | Which container has the highest initial rate of heat loss?
- A** P
B Q
C R
D S

- 10** A laser beam is reflected sequentially off two plane mirrors, M_1 and M_2 . The direction of the incident beam on M_1 is fixed. M_1 is rotated by an angle θ in a clockwise direction in the plane of incidence. To ensure that the final beam leaving M_2 maintains its original direction, by what angle and in which direction must M_2 be rotated?
- A** θ clockwise
 - B** θ anticlockwise
 - C** 2θ clockwise
 - D** 2θ anticlockwise
 - E** $\frac{\theta}{2}$ clockwise

- 11** A U-tube of uniform cross-sectional area, open to the atmosphere at both ends, initially contains a liquid of density ρ_1 .

A second immiscible liquid of density ρ_2 is poured into the left arm, forming a column of height h .

A third immiscible liquid of density ρ_3 is then poured into the right arm until the top liquid surfaces in both arms are at the exact same horizontal level, as shown in the diagram.

Given that $\rho_1 > \rho_2 > \rho_3$, which expression gives the height h_3 of the column of the third liquid?

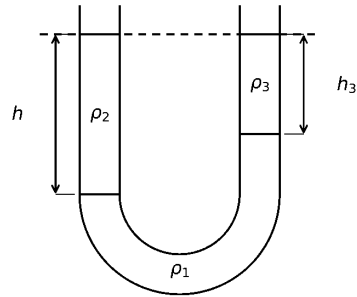


Diagram not to scale

- A** $\left(\frac{\rho_2}{\rho_3}\right) h$
- B** $\left(\frac{\rho_3}{\rho_2}\right) h$
- C** $\left(\frac{\rho_1 - \rho_2}{\rho_1 - \rho_3}\right) h$
- D** $\left(\frac{\rho_1 - \rho_3}{\rho_1 - \rho_2}\right) h$
- E** $\left(\frac{\rho_2 - \rho_3}{\rho_1 - \rho_3}\right) h$
- F** $\left(\frac{\rho_1 - \rho_2}{\rho_2 - \rho_3}\right) h$
- G** $\left(\frac{\rho_1 + \rho_2}{\rho_1 + \rho_3}\right) h$
- H** $\left(\frac{\rho_1 - \rho_2}{\rho_1 + \rho_3}\right) h$

- 12** Plane water waves travel along a straight canal. At a stationary post, wave crests are observed to pass every 2.0 s. A boat travels along the canal in the same direction as the waves with a constant speed of 4.0 m/s. An observer on the boat records that wave crests overtake the boat every 6.0 s.

What is the speed of the waves?

- A** 2.0 m/s
B 3.0 m/s
C 4.0 m/s
D 6.0 m/s
E 12.0 m/s
- 13** A small block of mass 2.0 kg is released from rest at the top of a smooth, curved ramp at a height $H = 20$ m above horizontal ground. The end of the ramp is horizontal and is at a height $h = 5.0$ m above the ground, where the block leaves the ramp.

Air resistance is negligible.

The gravitational field strength is $g = 10 \text{ N kg}^{-1}$.

What is the horizontal distance d travelled by the block between leaving the ramp and landing on the ground?

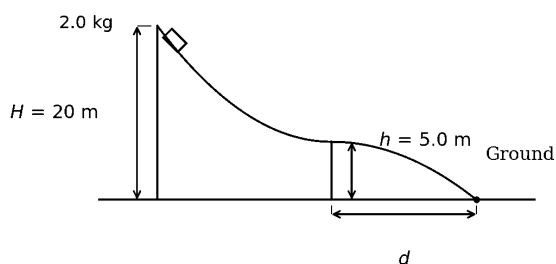


Diagram not to scale

- A** $5\sqrt{3}$ m
B 10 m
C $10\sqrt{3}$ m
D 15 m
E 20 m
F $20\sqrt{3}$ m

- 14** Three identical resistors are connected in a circuit with a DC power supply. One resistor is connected in series with a parallel combination of the other two.
What fraction of the total power supplied by the source is dissipated by the parallel combination?
- A** $\frac{1}{9}$
B $\frac{1}{6}$
C $\frac{1}{4}$
D $\frac{1}{3}$
E $\frac{1}{2}$
F $\frac{2}{3}$
- 15** An electric pump lifts oil of density 800 kg m^{-3} through a vertical height of 5.0 m at a constant volumetric flow rate of $2.5 \times 10^{-3} \text{ m}^3 \text{ s}^{-1}$. The pump system has an overall efficiency of 50% and is connected to a 50 V DC power supply.
Assuming $g = 10 \text{ m s}^{-2}$, what is the current drawn from the supply?
- A** 1.0 ext A
B 2.0 ext A
C 2.5 ext A
D 4.0 ext A
E 5.0 ext A
- 16** A continuous wave travels from medium P, where its speed is v , into medium Q, where its speed is $0.8v$.
Which of the following gives the fractional change in the wavelength of the wave?
- A** $-\frac{1}{4}$
B $-\frac{1}{5}$
C 0
D $\frac{1}{5}$
E $\frac{1}{4}$

- 17** A solid cuboid is made from a material of density 2000 kg m^{-3} . Its side lengths are in the ratio 1 : 2 : 4. The cuboid is placed on a horizontal surface to exert the maximum possible pressure. This pressure is 8.0 kPa.

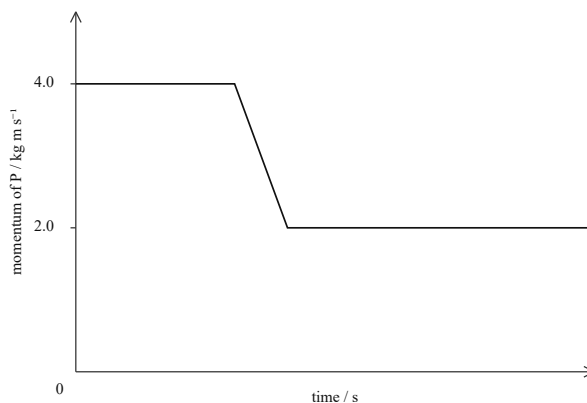
What is the mass of the cuboid?

(The gravitational field strength g is 10 N kg^{-1})

- A** 2.0 kg
 - B** 16 kg
 - C** 128 kg
 - D** 160 kg
 - E** 1024 kg
 - F** 16000 kg
- 18** A uniform ladder of mass M and length L rests against a smooth vertical wall and a rough horizontal floor with coefficient of static friction μ . A person of mass M climbs the ladder.
- Which of the following inequalities defines the range of angles θ (between the ladder and the floor) for which the ladder will remain stable for all positions of the person on the ladder?

- A** $\tan \theta \geq \frac{1}{4\mu}$
- B** $\tan \theta \geq \frac{1}{2\mu}$
- C** $\tan \theta \geq \frac{3}{4\mu}$
- D** $\tan \theta \geq \frac{1}{\mu}$

- 19** An object P collides with a stationary object Q on a straight, frictionless track. The graph shows the momentum of object P as a function of time.



What is the momentum of object Q after the collision?

- A** -4.0 kg m s^{-1}
 - B** -2.0 kg m s^{-1}
 - C** 0 kg m s^{-1}
 - D** 2.0 kg m s^{-1}
 - E** 4.0 kg m s^{-1}
 - F** 6.0 kg m s^{-1}
- 20** Four ions W, X, Y, and Z have the following compositions of protons (p), neutrons (n), and electrons (e):

	p	n	e
W	5	5	3
X	8	7	7
Y	10	10	9
Z	11	11	9

All four ions are accelerated to the same speed and enter a region of uniform magnetic field, with their velocity perpendicular to the field.

Which ion follows a circular path with the largest radius?

- A** Ion W
- B** Ion X
- C** Ion Y
- D** Ion Z
- E** All four ions have the same radius.
- F** It is not possible to determine without knowing the speed or magnetic field strength.

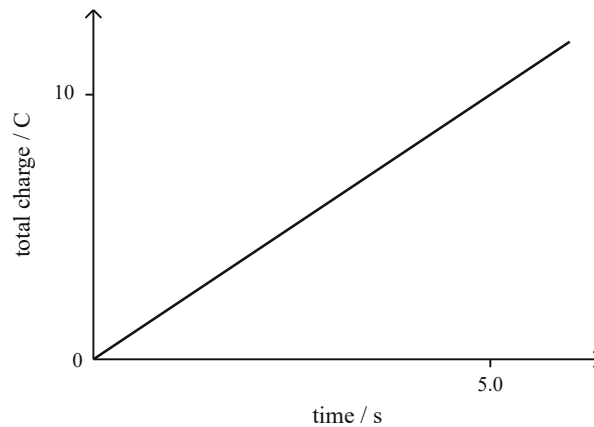
- 21** A large rigid tank filled with water is heated gently from the bottom. Consider a small parcel of water located immediately above the heat source. As the parcel absorbs thermal energy, it expands and begins to accelerate upwards.
- Which of the following statements best explains the physical cause of this initial upward acceleration?
- A** The density of the parcel decreases, so the gravitational force on the parcel becomes less than the upthrust exerted by the surrounding water.
 - B** The density of the parcel increases, so the upthrust exerted by the surrounding water becomes greater than the gravitational force on the parcel.
 - C** The internal pressure of the parcel increases, generating a net upward force that exceeds the gravitational force on the parcel.
 - D** The average kinetic energy of the molecules in the parcel increases, causing them to diffuse upwards against gravity more rapidly than they diffuse downwards.
 - E** The viscosity of the parcel decreases, reducing the drag force and allowing the surrounding pressure to push the parcel upwards.
- 22** An ideal gas bubble of volume V is released at a depth $3H$ below the surface of a liquid. It rises to a depth H .
The atmospheric pressure at the surface is equal to the pressure exerted by a column of the liquid of height H .
Assuming the temperature of the gas remains constant, what is the volume of the bubble at depth H ?
- A** $\frac{V}{3}$
 - B** $\frac{V}{2}$
 - C** $2V$
 - D** $3V$
 - E** $4V$
- 23** A laser beam from a fixed source is incident on a plane mirror M_1 . The reflected beam then strikes a second plane mirror M_2 . M_1 rotates with angular speed ω and M_2 rotates with angular speed 4ω . Both mirrors rotate in the same direction about parallel axes perpendicular to the plane of incidence.
- What is the angular speed of the final beam reflected from M_2 ?
- A** 3ω
 - B** 5ω
 - C** 6ω
 - D** 8ω
 - E** 10ω

- 24** A solid cuboid of mass 1.0 kg is placed on a horizontal surface. Depending on which face it rests on, it can exert one of three possible pressures: 0.40 N cm^{-2} , 0.50 N cm^{-2} , or 2.0 N cm^{-2} .

What is the density of the cuboid's material, in g cm^{-3} ?

(Take the gravitational field strength g to be 10 N kg^{-1} .)

- A** 0.05
B 0.40
C 2.0
D 2.5
E 20
F 20000
- 25** A resistor of resistance 4.0Ω is connected to a power supply. The graph shows the total charge that has passed through the resistor as a function of time.



What is the potential difference across the resistor?

- A** 0.50 V
B 2.0 V
C 8.0 V
D 20 V
E 25 V
F 40 V

- 26** A charged particle of mass m is released from rest in a uniform field. The particle experiences a constant force F due to the field.

What is the kinetic energy of the particle after it has travelled a distance d parallel to the force?

- A** $\frac{1}{2}Fd$
- B** $2Fd$
- C** Fd
- D** $\frac{Fd}{m}$
- E** Fdm

- 27** A set of six cylindrical wires, labelled A to F, are made from the same resistive material. Their lengths, L , and diameters, d , are given in the table.

Wire	Length L / m	Diameter d / mm
A	1.0	1.0
B	2.0	1.0
C	0.5	2.0
D	1.0	2.0
E	2.0	2.0
F	4.0	3.0

Each wire is connected in turn to the same power supply, which provides a constant voltage.

Which wire will carry the largest electric current?